TU-E2231 Machine Learning in Financial Risk Management
Spring 2025
Ruth Kaila and Eljas Toepfer

Instructions for Exercise 4

1. This exercise is graded passed/failed. The early-bird deadline is Sunday, **16.3.2025**. The final deadline is **16.4.2025**.

   To pass the course, the student must complete 6 weekly exercises. However, if the student returns each exercise within 10 days of the date the exercise was given, 5 exercises are sufficient to pass the course.

2. **IMPORTANT:** Return 2 or 3 pdf-files
   1. A PDF file with
      - I. A: Individual work (one page)
      - II. B: answers to the questions in the Python files
   2. A PDF file of the entire Python file 4.3 (see the instructions at the end for guidance on converting Python files to PDF)
   3. **For the 5 and 6 credit versions of the course:** A pdf-file of the Python Exercise 4.4 (S&P 500 index).

A. Written exercise (Individual work)

Write a total of one page (400-500 words) on the following 2 topics. Don't write about things that are already trivial to you. The aim of this exercise is to learn new things.

1. Outlier detection: write about the risks and information content of financial anomalies and outliers. Rather than asking ChatGPT, try to think yourself. You can comment on the following:
   - Outliers in Risk management
   - Fraud detection; examples
   - the role of outliers in Regulatory compliance
   - the risks of anomalies for Algo trading

2. Clustering: Describe the process of clustering financial assets based on historical price data. What are some common distance metrics or similarity measures used in this context, and how do they affect the clustering results? What kind of challenges and limitations are there when clustering financial data? What factors should be considered when interpreting the results of clustering analysis.

   Advanced: Provide examples of real-world applications of clustering, such as portfolio construction, risk management, or anomaly detection. How do clustering algorithms such as K-means, hierarchical clustering, or DBSCAN help solving these problems?

B. Python exercises

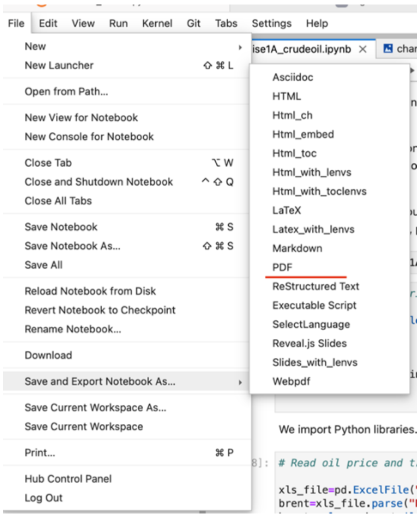
- Exercises 4.1 and 4.2 are based on de Prado’s book. In these exercises, the code is given. Read the code, make sure you understand the financial issue being discussed, and answer the questions on a separate sheet of paper. The code is given to you so that you have the energy to play with it. Change the parameters where possible and see how it affects the results/charts. This will improve your understanding of the theory presented.

- Exercise 4.3 is about fraud detection with unlabeled data. Read the code, answer the questions, and complete simple pieces of code. Pictures of some correct intermediary steps are at the end of this paper.

- **5/6 credit version of the course**: go through Exercise 4.4 on the S&P500 index, clustering and PCA. Answer carefully also the questions at the end of the file.

IMPORTANT: Focus on what you think is essential for you.

How to save the notebook as a pdf file:



4.3

[8]:

category	amount	fraud
es_barsandrestaurants	43.841793	0.022472
es_contents	55.170000	0.000000
es_fashion	59.780769	0.020619
es_food	35.216050	0.000000
es_health	126.604704	0.242798
es_home	120.688317	0.208333
es_hotelservices	172.756245	0.548387
es_hyper	46.788180	0.125000
es_leisure	229.757600	1.000000
es_otherservices	149.648960	0.600000
es_sportsandtoys	157.251737	0.657895
es_tech	132.852862	0.179487
es_transportation	27.422014	0.000000
es_travel	231.818656	0.944444
es_wellnessandbeauty	66.167078	0.060606

```
Flagged Fraud     0    1
Actual Fraud
0              6973   16
1               176   24
```